

LECTURE 3

Simple linear Regression

$$(x_1, y_1) \dots (x_n, y_n)$$

$$y_i \in \mathbb{R}$$

$$x_i \in \mathbb{R}$$

$$y_i = \underbrace{\beta_0 + \beta_1 x_i}_{\text{parameters}} + \underbrace{\varepsilon_i}_{\text{error}}$$

Statistical Inference: Frequentist
Bayesian

Frequentist Inference

parameters: θ in general
 (β_0, β_1)

data: $(x_1, y_1) \dots (x_n, y_n)$

- ① Construct some estimate of θ .
(e.g. least squares estimators)
Maximum likelihood estimate.

$$\varepsilon_1, \dots, \varepsilon_n \text{ i.i.d. } N(0, \sigma^2)$$

$$y_i = \beta_0 + \beta_1 x_i + \varepsilon_i$$

$$\text{Likelihood} = \prod_{i=1}^n \frac{1}{\sqrt{2\pi}\sigma} \exp\left[-\frac{(y_i - \beta_0 - \beta_1 x_i)^2}{2\sigma^2}\right]$$

$y_i \overset{\text{independent}}{\sim} N(\beta_0 + \beta_1 x_i, \sigma^2) \leftarrow$

Optimization (Maximization)

$$\text{Log-likelihood} = \sum_{i=1}^n \left[\log\left(\frac{1}{\sqrt{2\pi}\sigma}\right) - \frac{\log \sigma}{2} - \frac{(y_i - \beta_0 - \beta_1 x_i)^2}{2\sigma^2} \right]$$

$$= n \log \frac{1}{\sqrt{2\pi}} - n \log \sigma - \frac{\sum_{i=1}^n (y_i - \beta_0 - \beta_1 x_i)^2}{2\sigma^2}$$

$$S(\beta_0, \beta_1) = \sum_{i=1}^n (y_i - \beta_0 - \beta_1 x_i)^2$$

$$n \log \frac{1}{\sqrt{2\pi}} - n \log \sigma - \frac{S(\beta_0, \beta_1)}{2\sigma^2}$$

Check: $\hat{\beta}_0, \hat{\beta}_1$ coincide with the least squares estimators
MLE

$$\hat{\beta}_1 = \frac{\sum_{i=1}^n (y_i - \bar{y})(x_i - \bar{x})}{\sum_{i=1}^n (x_i - \bar{x})^2}$$

$$\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}$$

Check: $\hat{\sigma}_{MLE} = \sqrt{\frac{S(\hat{\beta}_0, \hat{\beta}_1)}{n}}$

② Characterize the distribution of the estimates.

$$\hat{\beta}_1 = \frac{\sum_{i=1}^n (y_i - \bar{y})(x_i - \bar{x})}{\sum_{i=1}^n (x_i - \bar{x})^2} \quad \begin{aligned} &= \frac{\sum_{i=1}^n (y_i - \bar{y})(x_i - \bar{x})}{\sum_{i=1}^n (x_i - \bar{x})^2} \\ &= \frac{\sum y_i (x_i - \bar{x})}{\sum \bar{y} (x_i - \bar{x})} \end{aligned}$$

$$y_i \text{ ind } N(\beta_0 + \beta_1 x_i, \sigma^2)$$

Assume x_i are fixed.

$$= \frac{\sum_{i=1}^n y_i (x_i - \bar{x})}{\sum_{i=1}^n (x_i - \bar{x})^2}$$

$$\sim N\left(\frac{\sum_{i=1}^n (\beta_0 + \beta_1 x_i) (x_i - \bar{x})}{\sum_{i=1}^n (x_i - \bar{x})^2}, \frac{\sigma^2 \sum_{i=1}^n (x_i - \bar{x})^2}{\left[\sum_{i=1}^n (x_i - \bar{x})^2\right]^2}\right)$$

$$= N\left(\beta_1, \frac{\sigma^2}{\sum_{i=1}^n (x_i - \bar{x})^2}\right)$$

$$\hat{\beta}_1 \sim N\left(\beta_1, \frac{\sigma^2}{\sum_{i=1}^n (x_i - \bar{x})^2}\right)$$

$$\hat{\sigma}_{MLE}^2 = \frac{S(\hat{\beta}_0, \hat{\beta}_1)}{n}$$

$$n \frac{\hat{\sigma}_{MLE}^2}{\sigma^2} \sim \chi_{n-2}^2$$

Chi-squared distribution with $n-2$ degrees of freedom

Mean of $\chi_{n-2}^2 = n-2$

$$E \hat{\sigma}_{MLE}^2 = \sigma^2 \frac{n-2}{n}$$

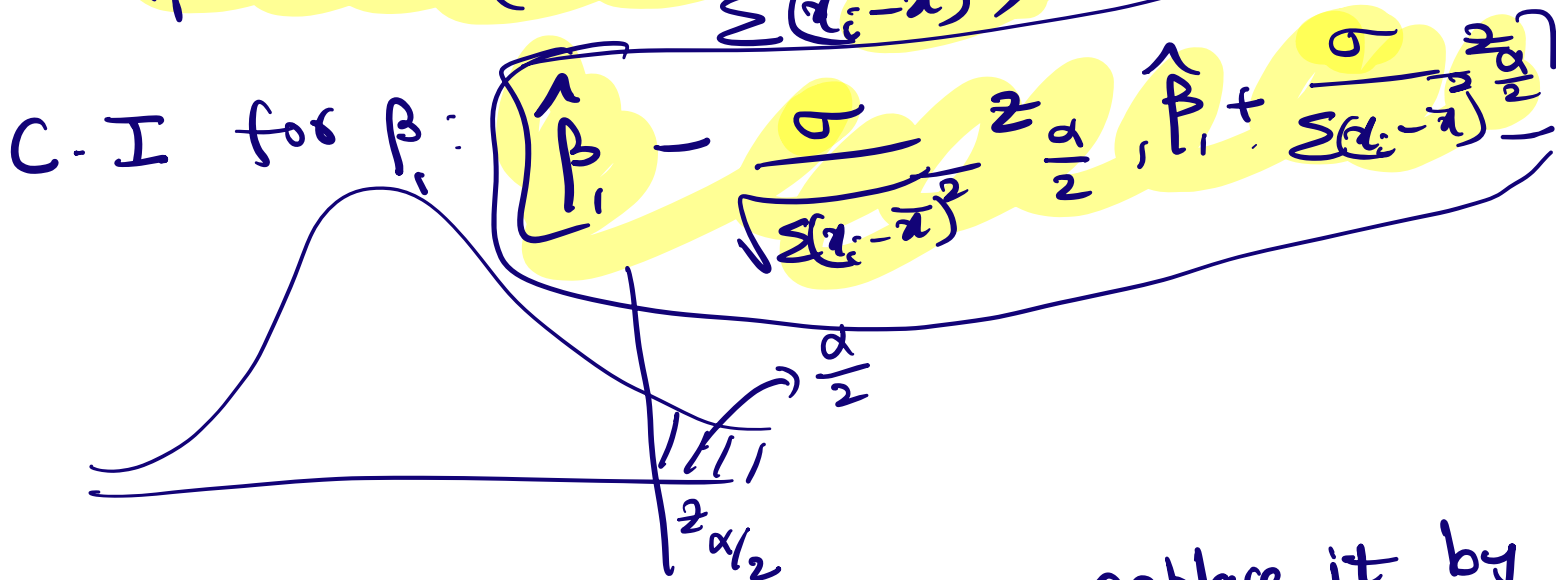
$\hat{\sigma}_{MLE}^2$: biased estimator

$$\hat{\sigma}_{unbiased}^2 = \frac{n}{n-2} \hat{\sigma}_{MLE}^2 = \frac{1}{n-2} S(\hat{\beta}_0, \hat{\beta}_1)$$

If we do Auto Regression:

$$\hat{\beta}_1 = \frac{\sum (y_i - \bar{y})(y_{i-1} - \bar{y})}{\sum (y_{i-1} - \bar{y})^2}$$

③ $\hat{\beta}_1 \sim N\left(\beta_1, \frac{\sigma^2}{\sum (x_i - \bar{x})^2}\right)$



Because σ is unknown, you replace it by $\hat{\sigma}^2_{\text{unbiased}}$

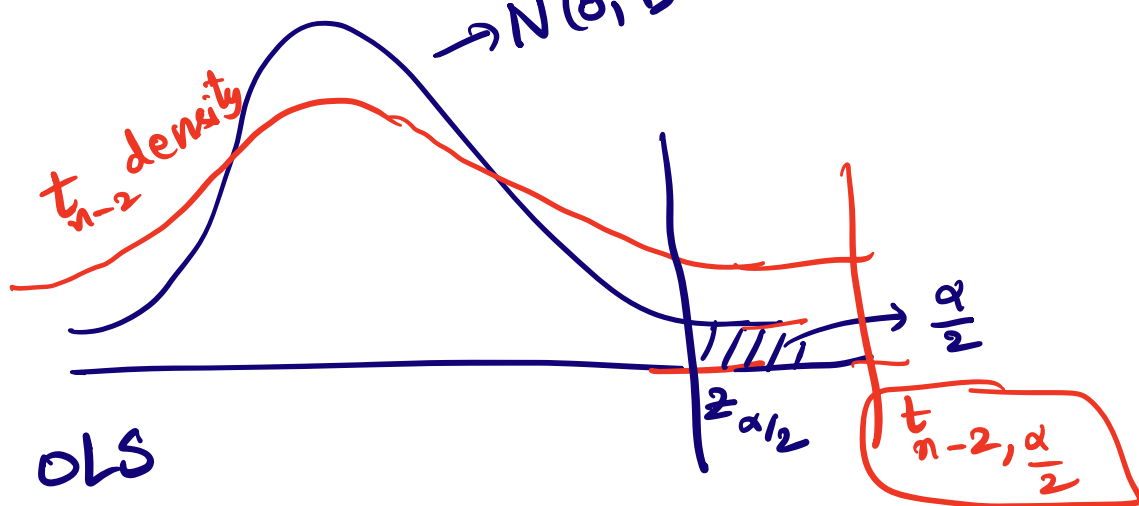
$$\hat{\beta}_1 \sim N\left(\beta_1, \frac{\sigma^2}{\sum (x_i - \bar{x})^2}\right)$$

$$\frac{\hat{\beta}_1 - \beta_1}{\hat{\sigma}_{\text{unbiased}}} \sqrt{\sum (x_i - \bar{x})^2} \sim t_{n-2}$$

Studentized $\hat{\beta}_1$

Student t-distribution with $n-2$ degrees of freedom

sm. ols



<u>coef</u>	<u>b.e</u>	<u>t</u>	<u>[]</u>
*	*	*	*

$\sigma \rightarrow \hat{\sigma}$
unbiased

$z_{\frac{\alpha}{2}} \rightarrow t_{n-2, \frac{\alpha}{2}}$

$$Y_1 \sim N(Y_2, 1)$$

Bayesian Inference.

$$Y_1 - Y_2 \sim N(0, 1)$$

$$y_i = \beta_0 + \beta_1 x_i + \varepsilon_i, \quad \varepsilon_i \stackrel{i.i.d}{\sim} N(0, \sigma^2)$$

$$y_i \stackrel{\text{independent}}{\sim} N(\beta_0 + \beta_1 x_i, \sigma^2)$$

β_0, β_1, σ

① Select a probability distribution for the parameters (PRIOR)

— Captures your information about the parameters before you see the data

② Write likelihood & treat the likelihood as the conditional density of the data given the parameters.

③ Use the rules of probability (Bayes Rule) to compute the conditional distribution of parameters given observed data.

④ Use the posterior distribution to infer the parameters. → POSTERIOR DISTRIBUTION

Back to linear regression

$$\beta_0, \beta_1, \sigma \sim$$

$\beta_0, \beta_1, \log \sigma \stackrel{\text{iid}}{\sim} \text{Unif}[-C, C]$ for a large C

$$\log \sigma \sim \text{Unif}[-C, C]$$

$$\Rightarrow f_{\log \sigma}(x) = \frac{\mathbb{I}\{-C < x < C\}}{2C}$$

$$\begin{aligned} \Rightarrow f_{\sigma}(u) &= f_{\log \sigma}(\log u) \times \frac{1}{u} \\ &= \frac{\mathbb{I}\{-C < \log u < C\}}{2C} \\ &= \frac{\mathbb{I}\{e^{-C} < u < e^C\}}{2Cu} \end{aligned}$$

$$f_{\sigma}(\sigma) \propto \frac{1}{\sigma}$$

$$f\left(\frac{t}{2}\right) = f_{\sigma}\left(\frac{t}{2}\right) \times \frac{1}{2}$$

$$\frac{1}{2\sigma} = \frac{2}{t} \times \frac{1}{2} = \frac{1}{t}$$

$$\beta_0, \beta_1, \log \sigma \text{ iid Unif}(-C, C)$$

$$f(\beta_0, \beta_1, \sigma) = \frac{\mathbb{I}\{-C < \beta_0, \beta_1, \log \sigma < C\}}{2C \times 2C \times 2C \times \sigma}$$

$$\propto \frac{1}{\sigma} \mathbb{I}\{-C < \beta_0, \beta_1, \log \sigma < C\}$$

likelihood:

$$f(\text{data} | \beta_0, \beta_1, \sigma) = \prod_{i=1}^n \frac{1}{\sqrt{2\pi} \sigma} \exp\left(-\frac{(y_i - \beta_0 - \beta_1 x_i)^2}{2\sigma^2}\right)$$

$$\propto \prod_{i=1}^n \left[\frac{1}{\sigma} \exp\left(-\frac{(y_i - \beta_0 - \beta_1 x_i)^2}{2\sigma^2}\right) \right]$$

$$= \sigma^{-n} \exp\left[-\frac{S(\beta_0, \beta_1)}{2\sigma^2}\right]$$

$$f_{\beta_0, \beta_1, \sigma} \quad , \quad f_{\text{data} | \beta_0, \beta_1, \sigma} \rightarrow f_{\beta_0, \beta_1, \sigma | \text{data}}$$

$\downarrow$ PRIOR $\downarrow$ likelihood

BAYES RULE: How to go from $P(A|B)$ to $P(B|A)$?

e.g. A : someone died
B : they were hanged

$$P(A|B) = 1$$

$$P(B|A) \approx 0$$

$$P(B|A) = \frac{P(A|B) P(B)}{P(A)}$$

$$P(B|A) = \frac{P(A|B) P(B)}{P(A|B) P(B) + P(A|B^c) P(B^c)}$$

$$f(\text{parameters} | \text{data}) = \frac{f(\text{data} | \text{parameters}) f(\text{parameters})}{f(\text{data})}$$

$$f(\text{data}) = \frac{f(\text{data} | \text{parameters}) f(\text{parameters})}{\int f(\text{data} | \text{parameters}) f(\text{parameters}) d(\text{parameters})}$$

Simplified Bayes Rule:

$$\underbrace{f(\text{parameters} | \text{data})}_{\text{Posterior}} \propto \underbrace{f(\text{data} | \text{parameters})}_{\text{Likelihood}} \underbrace{f(\text{parameters})}_{\text{Prior}}$$

$$f(\beta_0, \beta_1, \sigma | \text{data}) \propto \frac{1}{\sigma^n} \exp\left(-\frac{S(\beta_0, \beta_1)}{2\sigma^2}\right) \frac{1}{\sigma} I\left\{-C < \beta_0, \beta_1, \text{ or } \sigma < C\right\}$$

posterior $\propto \sigma^{n-1} \exp\left(-\frac{S(\beta_0, \beta_1)}{2\sigma^2}\right) I\left\{-C < \beta_0, \beta_1, \text{ or } \sigma < C\right\}$

$$\boxed{\beta_0, \beta_1}$$

$$x_1, x_2, x_3 \quad \int_{x_1, x_2, x_3} f(x_1, x_2, x_3) \rightarrow \text{INTEGRATION}$$

$$\text{Integration: } \int \beta_0, \beta_1 | \text{data}$$